Technical Data Sheet

Retrofit Absorber – makes your PV-panel produce both photovoltaic and thermal energy

Copa Solar

Article No
RAM00203

General description

RAM00203 is a Thermal Heat Absorber plate for Retrofit on existing PV installations. The Retrofit Absorber consists of aluminium pipes on an aluminium cooling plate, prepared with adhesive stripes for mounting on the back side of a PV panel.

The Retrofit Absorber is designed to lead away excess heat that is built up in a PV panel and its surroundings from the solar irradiance and make the PV panel work at its optimal temperature.

The thermal energy can be used in various applications, and the electrical energy production will increase due to the optimization of the panel´s operating temperature.

General data

Physical properties - Retrofit absorber			
Dimensions outer edge	L x W x H	1074 x 1662 x 13	mm
Total Area outer edge		1,79	m^2
Weight	Empty │ Full	3,9 │ 5,7	kg

Physical requirements – PV panel			
Dimensions outer edge	L x W	min 1722 x min 1134	mm
Dimensions frame opening (back)	L x W	min 1074 x min 1662	mm

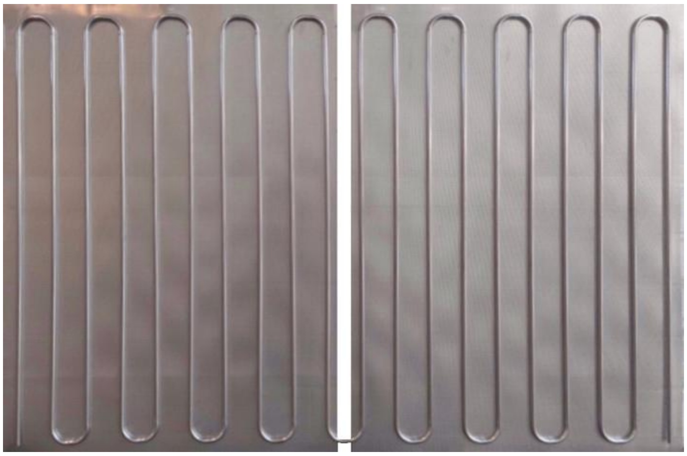
Guidelines and preparations

Individual assessments in collaboration with Copa Solar are needed in each case.

The mounting of a retrofit absorber needs to made according to the Copa Solar Guidelines, taking outdoor temperature and humidity in consideration.

The back surface of the PV panels needs to be prepared according to the Copa Solar Guidelines for Retrofit on existing PV installations.

Detail



Thermal data

Collector characteristics			
Thermal Power Output	Peak	2500 W_{th}/abs	1430 W_{th}/m^2
Thermal Power Output	ΔT=0	925 W_{th}/abs	530 W_{th}/m^2
Optical Efficiency	η_0	0,53	
Heat Loss Coeff 1st order	a_1	22,5	W/m²·K
Heat Loss Coeff 2nd order	a_2	0	W/m²·K²
Absorber plate dimension	L x W	1074 x 816	mm
Active thermal area	$A_{absorber}$	1,76	m^2
Heat Transfer Material (Alu)	plate │ pipe	EN AW-1050A │ EN AW-3103	

Thermal Power Output Peak is calculated at T_m-T_a=-40 K, G=1000 W/m²

Heat transfer characteristics			
Thermal Fluid Volume		1,73 L	
Max Operation Pressure		15 bar	
Stagnation Temperature		80°C	
Hydraulic Connection		12 mm Pushfit	
Pressure Loss	at 120 L/h	8,6 kPa	0,83 mH_2O
	at 180 L/h	13,1 kPa	1,27 mH_2O

Pressure Loss is calculated at T=20°C, Heat Transfer Fluid: Water/Glykol 60/40

Copa Solar AB
www.copasolar.com
Stålgatan 15A
SE-195 72 Rosersberg, SWEDEN
Phone: +46-70-822 17 20
E-mail: info@copasolar.com

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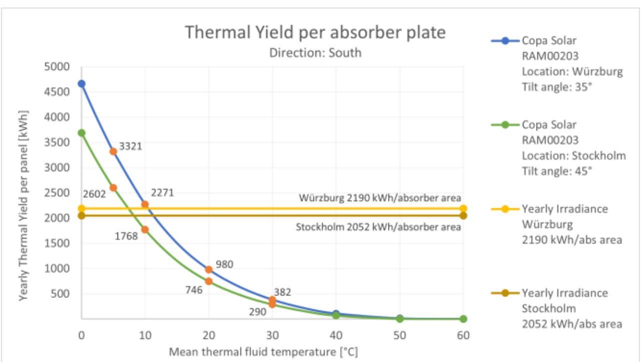
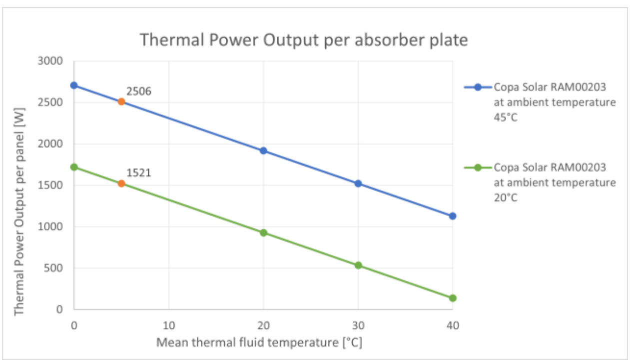
Yearly Thermal Yield

Calculation of Thermal Yield is made with ScenoCalc v6.2.



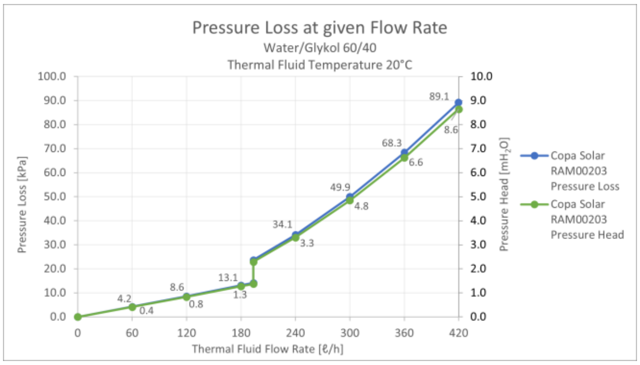
Thermal Power Output

Thermal Power Output is evaluated in relation to ambient temperature at 45°C and 20°C.
The performance of the thermal fluid is best at low mean temperatures.



Pressure drop

The pressure drop and pressure head are calculated with the pipe characteristics.
The laminar–turbulent transition occurs at ~193 L/h.



Additional information

Conforms to the following standards

EN 12975:2022	SS EN 61215-1:2021
EN 62938	SS EN 61215-2:2021
EN 61730-1	SS EN 61215-1-1:2021
	SS EN 61215-1-2:2021
	SS EN 61215-1-3:2021
	SS EN 61215-1-4:2021

UN CPC code
173 Steam and hot water
NACE Code
D351133 Alternative and renewable sources of energy: solar
GPC code
10005875 Solar Power Stations

Model No
RAM-M-ALAL-1017-22A

Subject to technical modifications and any changes.

Copa Solar AB
www.copasolar.com
Stålgatan 15A
SE-195 72 Rosersberg, SWEDEN
Phone: +46-70-822 17 20
E-mail: info@copasolar.com